Stat 145 (Multivariate Statistics)

Laboratory Exercise No. 1

INSTRUCTIONS: Show a detailed and neat solution.

1. Let $\boldsymbol{X} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with

$$\boldsymbol{\mu} = \begin{bmatrix} -3 \\ 1 \\ 4 \end{bmatrix}$$

and

$$\boldsymbol{\Sigma} = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

   a) Find the distribution of $W_1 = 3x_1 - x_2 + 2x_3 + 1$.

   b) Find the distribution of $W_2 = x_2 - 3x_3 + 5$.

   c) Find the joint distribution of $\boldsymbol{W} = \begin{bmatrix} W_1 \\ W_2 \end{bmatrix}$. Is W_1 independent of W_2?

   d) Is X_1 independent of $W_3 = x_1 + 3x_2 - 2x_3$?

2. Consider $\boldsymbol{Y} \sim N_5(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\mu}' = \begin{bmatrix} 1 & -1 & 2 & 4 & 0 \end{bmatrix}$$

and

$$\boldsymbol{\Sigma} = \begin{bmatrix} 4 & 0 & -1 & 2 & 3 \\ 0 & 5 & 0 & 2 & -1 \\ -1 & 0 & 2 & 3 & 4 \\ 2 & 2 & 3 & 4 & 0 \\ 3 & -1 & 4 & 0 & 1 \end{bmatrix}$$

Find the conditional distribution of $\boldsymbol{Y}_1' = \begin{bmatrix} Y_1 & Y_3 & Y_5 \end{bmatrix}$ given $\boldsymbol{Y}_2' = \begin{bmatrix} Y_2 = 3 & Y_4 = -1 \end{bmatrix}$.